

Data Mining: Concepts and Techniques

1 What is Data Mining?

Data Mining

Data mining is the process of discovering **patterns**, **anomalies**, and **useful insights** from large datasets using techniques from **machine learning**, **statistics**, and **database systems**.

It involves transforming raw data into actionable information for better decision-making, prediction, and strategic advantage. Analysts often use the analogy of finding valuable **"gold"** (**insights**) within massive amounts of **"ore"** (**data**).

2 How It Works: The Workflow

1. **Data Collection & Preparation:** Gathering data from various sources, cleaning it, and loading it into a warehouse (ETL process).
2. **Pattern Discovery:** Applying algorithms (often AI/ML) to find relationships, trends, and correlations.
3. **Evaluation:** Assessing the discovered patterns to ensure they are meaningful, valid, and useful.
4. **Action/Deployment:** Using the insights for tasks like predicting customer needs, detecting fraud, or optimizing operations.

3 Key Techniques & Real-World Examples

- **Classification:** Categorizing data into predefined classes (e.g., spam vs. not spam).
- **Clustering:** Grouping similar data points together without pre-defined labels.
- **Association Rule Mining:** Finding items that often occur together (e.g., "Basket Analysis").
- **Regression:** Predicting numerical values based on historical data.

Applications

- **Retail:** Personalized product recommendations (Amazon).
- **Finance:** Detecting fraudulent credit card transactions.
- **Healthcare:** Identifying patients at high risk for certain diseases.
- **Marketing:** Targeting specific customer segments for campaigns.

4 Deep-Dive: Association Rule Mining (Apriori)

4.1 Problem Statement

Given the following dataset of 5 transactions, discover strong association rules using the **Apriori Algorithm**.

- **Thresholds:** $min_support = 40\%$ (count ≥ 2), $min_confidence = 70\%$.

TID	Items Bought
1	{Bread, Butter, Milk}
2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Beer, Diapers}

4.2 Step 1: Frequent Itemset Generation

The Apriori process involves iteratively generating **Candidate sets** (C_k) and pruning them into **Frequent sets** (L_k) until no more itemsets can be formed.

Pass 1: Identifying Frequent 1-itemsets (L_1)

Candidate set C_1			Frequent set L_1	
Itemset	Count		Itemset	Count
Bread	3	$\xrightarrow{min_sup=2}$	Bread	3
Butter	3		Butter	3
Milk	2		Diapers	3
Beer	2		Milk	2
Cookies	1		Beer	2
Diapers	3			

Pass 2: Identifying Frequent 2-itemsets (L_2)

Candidate set C_2			Frequent set L_2	
Itemset	Count		Itemset	Count
{Bread, Butter}	3	$\xrightarrow{min_sup=2}$	{Bread, Butter}	3
{Bread, Diapers}	1		{Bread, Milk}	2
{Bread, Milk}	2		{Butter, Milk}	2
{Bread, Beer}	0		{Diapers, Beer}	2
{Butter, Diapers}	1			
{Butter, Milk}	2			
{Butter, Beer}	0			
{Diapers, Milk}	1			
{Diapers, Beer}	2			
{Milk, Beer}	0			

Pass 3: Identifying Frequent 3-itemsets (L_3)

Candidate set C_3		$\xrightarrow{\text{min_sup}=2}$	Frequent set L_3	
Itemset	Count		Itemset	Count
{Bread, Butter, Milk}	2		{Bread, Butter, Milk}	2
{Bread, Butter, Diapers}	1			
... (others are 0)				

4.3 Step 2: Rule Generation and Confidence Check

$\text{min_confidence} = 70\%$. Confidence ($X \rightarrow Y$) = $\frac{\text{Support_Count}(X \cup Y)}{\text{Support_Count}(X)}$.

Final Confidence Calculations and Strong Rules

- For {Bread, Butter}:
 - Bread $\rightarrow$ Butter: $3/3 = 100\%$ (**Strong**)
 - Butter $\rightarrow$ Bread: $3/3 = 100\%$ (**Strong**)
- For {Bread, Milk}:
 - Bread $\rightarrow$ Milk: $2/3 = 67\%$ (Weak)
 - Milk $\rightarrow$ Bread: $2/2 = 100\%$ (**Strong**)
- For {Butter, Milk}:
 - Butter $\rightarrow$ Milk: $2/3 = 67\%$ (Weak)
 - Milk $\rightarrow$ Butter: $2/2 = 100\%$ (**Strong**)
- For {Diapers, Beer}:
 - Diapers $\rightarrow$ Beer: $2/3 = 67\%$ (Weak)
 - Beer $\rightarrow$ Diapers: $2/2 = 100\%$ (**Strong**)
- For {Bread, Butter, Milk}:
 - Bread, Butter $\rightarrow$ Milk: $2/3 = 67\%$ (Weak)
 - Bread, Milk $\rightarrow$ Butter: $2/2 = 100\%$ (**Strong**)
 - Milk, Butter $\rightarrow$ Bread: $2/2 = 100\%$ (**Strong**)
 - Bread $\rightarrow$ Butter, Milk: $2/3 = 67\%$ (Weak)
 - Butter $\rightarrow$ Bread, Milk: $2/3 = 67\%$ (Weak)
 - Milk $\rightarrow$ Bread, Butter: $2/2 = 100\%$ (**Strong**)

Strong association rules reveal the hidden relationships within consumer data.